Group II, Row IV in your form: fragmentation function and DGLAP

Throughout:

- the phase $e^{-ik\cdot(y'-y)/\xi}$, the measure $\frac{d\xi}{\xi^2}$ and the $\frac{1}{k^+}$ are kept explicit;
- $d_\perp = 2-\epsilon$, and the collinear pole is infrared ($\epsilon<0$);
- $\overline{\rm MS}$ subtracts the pole $-\frac{2}{\epsilon}$ (in my earlier notes $-\frac{2}{\epsilon}\equiv\frac{1}{\bar\epsilon}$);
- $X\equiv\frac{x-y}{(x-y)^2}$, $X'\equiv\frac{x'-y'}{(x'-y')^2}$, $\hat\Delta\equiv\frac{y'-y}{|y'-y|}$, and the colour factor is

$$\mathcal{O}\equiv\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x).$$

1 Your form

$$\begin{aligned}\frac{d^3N_{\rm IV}}{d^3k} &= \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,W\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big]\right]\mathcal{O}\\
&+\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,W\,\mathcal{O}\\
&+\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\\
&\quad\times\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-d_\perp\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big]\mathcal{O},\end{aligned}\tag{1.1}$$

with the weight

$$W\equiv-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}-1,\tag{1.2}$$

and your subtraction at $\xi=1$:

$$\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[\cdots+\frac{\xi}{(1-\xi)_+}\Big]\equiv\int_{\frac{k^+}{\vee}}^1 d\xi\Big\{\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}+\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}\Big]\frac{1}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big\}.\tag{1.3}$$

- **The $\delta(1-\xi)$ coefficient is exact:**

$$\int_{k^+/\vee}^{k^+/(k^++\Lambda)}\frac{d\xi}{\xi(1-\xi)}=\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\qquad(\texttt{sympy}:$$

  0).

- **The upper limit 1** instead of $\frac{k^+}{k^++\Lambda}$ in the first line adds the strip $\frac{k^+}{k^++\Lambda}<\xi<1$, i.e. $\hat R=-i\frac{\Lambda}{k^+}k\cdot(y'-y)\,e^{-ik\cdot(y'-y)}+O(\frac{\Lambda^2}{k^{+2}})$ inside the braces (times W and the colour structure).
  - With upper limit $\frac{k^+}{k^++\Lambda}$, Eq. (1.1) equals Row IV to rel. 2×10^{-14}.
  - With upper limit 1, at $\frac{\Lambda}{k^+}=10^{-4}$, the difference is 1.3×10^{-4}.

  Below, $\Lambda\to0$ is taken only in such power terms. $\log\frac{\vee}{\Lambda}$ is kept.

2 Step 1: pole and finite part of the weight

With $d_\perp = 2-\epsilon$, $-\frac{d_\perp}{\epsilon} = -\frac{2}{\epsilon}+1$. The $+1$ cancels the -1 of W:

$$W = -\frac{2}{\epsilon} - 2\gamma_E - \log\frac{(y'-y)^2\mu_{\overline{MS}}^2}{4} + O(\epsilon) \tag{2.1}$$

(`sympy`: difference 0). For $d_\perp = 2$ instead, $W = -\frac{2}{\epsilon} - 2\gamma_E - \log\frac{(y'-y)^2\mu^2}{4} - 1$. The pole is the same, and the finite part changes by a scheme term.

- The pole of Row IV is the first two lines of Eq. (1.1) with $W \to -\frac{2}{\epsilon}$.
- The third line has no pole.
- With $\frac{g^2 N_c}{4\pi^2} = \frac{\alpha_s}{2\pi}\cdot 2N_c$ (`sympy`: 0), the pole is

$$\begin{aligned}\left.\frac{d^3N_{\rm IV}}{d^3k}\right|_{\rm pole} &= \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}\\ &\quad\times 2N_c\Bigg\{\int_{\frac{k^+}{\vee}}^1 d\xi\Big[\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}+\Big(\xi(1-\xi)+\frac{1-\xi}{\xi}\Big)\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}}{\xi^2}\Big]\\ &\quad+\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]e^{-ik\cdot(y'-y)}\Bigg\}.\end{aligned} \tag{2.2}$$

3 Step 2: from your subtraction to the standard plus distribution

Why this step is needed. A fragmentation function $D(\xi,\mu)$ and its DGLAP kernel depend on ξ only.

- In your split, the subtraction term of Eq. (1.3) is $e^{-ik\cdot(y'-y)}\int_{k^+/\vee}^1\frac{d\xi}{\xi(1-\xi)}$. Its $\frac{1}{\xi}$ part gives $\int_{k^+/\vee}^1\frac{d\xi}{\xi}=\log\frac{\vee}{k^+}$, which depends on the lower limit $\frac{k^+}{\vee}$.
- Correspondingly, your $\delta(1-\xi)$ coefficient contains $\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})=\log\frac{\vee-k^+}{k^+}$, which depends on $k^+/\vee$.
- In the convolution with D the gluon momentum is $\frac{k^+}{\zeta}$, so the lower limit becomes $\frac{k^+}{\zeta\vee}$, and this piece changes with ζ. It therefore cannot be part of a kernel $P(\xi)$.

The standard plus distribution on $[0,1]$ subtracts only $\frac{e^{-ik\cdot(y'-y)}}{1-\xi}$:

$$\begin{aligned}\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,e^{-ik\cdot(y'-y)\frac{1}{\xi}} &\equiv 2N_c\Bigg\{\int_{\frac{k^+}{\vee}}^1 d\xi\Big[\frac{\frac{1}{\xi}e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}+\Big(\xi(1-\xi)+\frac{1-\xi}{\xi}\Big)\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}}{\xi^2}\Big]\\ &\quad+\log\Big(1-\frac{k^+}{\vee}\Big)e^{-ik\cdot(y'-y)}\Bigg\},\end{aligned} \tag{3.1}$$

with $P^{(r)}_{gg,+}(\xi)=2N_c\big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\big]$ and $\frac{\xi}{(1-\xi)_+}\equiv\xi\big[\frac{1}{1-\xi}\big]_+$. The $\log(1-\frac{k^+}{\vee})$ is produced by the plus distribution itself, because the integration starts at $\frac{k^+}{\vee}$.

The conversion (exact). The two integrands differ by

$$\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}-\frac{\frac{1}{\xi}e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}=-e^{-ik\cdot(y'-y)}\Big[\frac{1}{\xi(1-\xi)}-\frac{1}{1-\xi}\Big]=-\frac{e^{-ik\cdot(y'-y)}}{\xi},$$

and $\int_{k^+/\vee}^1\frac{d\xi}{\xi}=\log\frac{\vee}{k^+}$. Hence

$$\begin{aligned}&\int_{\frac{k^+}{\vee}}^1 d\xi\Big[\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}+\Big(\xi(1-\xi)+\frac{1-\xi}{\xi}\Big)\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}}{\xi^2}\Big]+\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]e^{-ik\cdot(y'-y)}\\ &\quad=\frac{1}{2N_c}\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}+\Big[\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}\Big]e^{-ik\cdot(y'-y)}\end{aligned} \tag{3.2}$$

(numerically, 20 random cases, rel. 1×10^{-15}). The shift between the two splits is $e^{-ik\cdot(y'-y)}\big[\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})\big]$ (`sympy`: twice it is $2\log\frac{\vee}{k^+}+2\log(1-\frac{k^+}{\vee})$). It moves from your $\delta(1-\xi)$ term into the plus integral.

The pole of Row IV in standard form.

$$\boxed{\left.\frac{d^3N_{\rm IV}}{d^3k}\right|_{\rm pole} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O} \\ \times\left\{\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,e^{-ik\cdot(y'-y)\frac{1}{\xi}} + 2N_c\Big[\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}\Big]e^{-ik\cdot(y'-y)}\right\}} \tag{3.3}$$

- The first term is the real part of P_{gg}, convolved with the phase $e^{-ik\cdot(y'-y)\frac{1}{\xi}}$ and the measure $\frac{d\xi}{\xi^2}$.
- The second is the soft logarithm $2N_c\log\frac{k^+}{\Lambda}$ at $\xi=1$ (phase without $\frac{1}{\xi}$).

4 Step 3: the $\delta(1-\xi)$ terms with Group I

Terms with $e^{-ik\cdot(y'-y)}$ (no $\frac{1}{\xi}$) and the LO structure also come from Group I: Row II and Row V (2b, UV), each with its c.c. There is also the charge renormalization of the LO coupling, $\alpha_0=\alpha_s(\mu)\big[1+\frac{\alpha_s\beta_0}{4\pi}(-\frac{2}{\epsilon})\big]$, with $\beta_0=\frac{11}{3}N_c$. For $\Lambda\to 0$ in power terms, the coefficients of

$$\frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)N_c\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}$$

are:

source	weight	coefficient
Row IV, Eq. (3.3) (standard split)	1	$2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}$
(in your split, Eq. (2.2))		$(2\log\frac{\vee}{\Lambda}+2\log(1-\frac{k^+}{\vee})$, with the shift in the plus integral)
Group I, Row II	$\times2$	$-\frac{11}{3}+2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}-\log\big(1-\frac{k^+}{\vee}\big)$
Group I, Row V 2b UV	$\times2$	$\frac{11}{3}-4\log\frac{\vee}{\Lambda}+4\log\frac{\vee}{k^+}$
charge renormalization		$+\frac{11}{6}$
sum		$\frac{11}{6}-\log\big(1-\frac{k^+}{\vee}\big)$ (`sympy`: $\frac{11}{6}-\log(1-\xi_0)$)

- $\log\frac{\vee}{\Lambda}$: $+2+2-4=0$, and $\log\frac{\vee}{k^+}$: $-2-2+4=0$. The soft logarithms cancel, as they must.
- The $\frac{11}{3}$ cancel between Row II and Row V 2b.
- What is left, $\frac{11N_c}{6}=\frac{\beta_0}{2}$, is the $\delta(1-\xi)$ coefficient of P_{gg}. The momentum sum rule $\int_0^1 d\xi\,\xi P_{gg}=0$ gives the same value (`sympy`: $\frac{11N_c}{6}$).
- **Remainder:** $-N_c\log(1-\frac{k^+}{\vee})$, of order $\frac{k^+}{\vee}$. It depends on $k^+/\vee$, not on ξ alone, so it cannot go into D. It must cancel against other rows, or the upper limits $\vee$ of Row II and Row IV must be made consistent; this is still open.

So, up to the remainder, the full pole of the gluon spectrum is

$$\left.\frac{d^3N_g}{d^3k}\right|_{\rm pole} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}, \tag{4.1}$$

$$P_{gg}(\xi)=2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]+\frac{11N_c}{6}\,\delta(1-\xi)\qquad(n_f=0). \tag{4.2}$$

The $\delta(1-\xi)$ term sets $\xi=1$ in the phase and in $\frac{1}{\xi^2}$.

5 Step 4: the fragmentation function

Factorization, written out. A final state h with $(k^+,\mathbf{k})$ comes from a gluon with $(q^+,\mathbf{q})=(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi})$:

$$\frac{d^3N_h}{d^3k}=\int_{\frac{k^+}{\vee}}^1 d\xi\,D^{(0)}_{h/g}(\xi)\int dq^+d^2\mathbf{q}\,\delta(k^+-\xi q^+)\,\delta^{(2)}(\mathbf{k}-\xi\mathbf{q})\,\frac{d^3N_g}{d^3q}(q^+,\mathbf{q}).$$

At LO, $\frac{d^3N_g}{d^3q}=\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2q^+}\int e^{-i\mathbf{q}\cdot(y'-y)}X\cdot X'\,\mathcal{O}$. Under the substitution:

- the δ functions give $\frac{1}{\xi^3}$;

- $\frac{1}{q^+} = \frac{\xi}{k^+}$;
- $e^{-i\mathbf{q}\cdot(y'-y)} = e^{-ik\cdot(y'-y)\frac{1}{\xi}}$.

So

$$\frac{d^3N_h^{\rm LO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{\frac{k^+}{\vee}}^1 \frac{d\xi}{\xi^2} D_{h/g}^{(0)}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}. \tag{5.1}$$

This has the same $\frac{d\xi}{\xi^2}$, phase and $\frac{1}{k^+}$ as Eq. (4.1).

The counterterm.

$$\boxed{\begin{aligned}&\Gamma_{gg}(z) = \delta(1-z) + \frac{\alpha_s}{2\pi}\frac{2}{\epsilon}P_{gg}(z),\\ &D_{h/g}^{(0)}(\xi) = \int_\xi^1 \frac{dz}{z}\Gamma_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu\Big) = D_{h/g}(\xi,\mu) + \frac{\alpha_s}{2\pi}\frac{2}{\epsilon}\int_\xi^1\frac{dz}{z}P_{gg}(z)D_{h/g}\Big(\frac{\xi}{z},\mu\Big).\end{aligned}} \tag{5.2}$$

At $O(\alpha_s)$ there are two pole terms.

- **A: Eq. (5.2) inserted into Eq. (5.1):**

$$A = \frac{\alpha_s}{2\pi}\frac{2}{\epsilon}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\Big[\int_\xi^1\frac{dz}{z}P_{gg}(z)D_{h/g}\Big(\frac{\xi}{z},\mu\Big)\Big]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}.$$

- **B: the pole (4.1), at the gluon momentum $(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$, fragmented with $\int\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta,\mu)$.** In Eq. (4.1):
  - $\frac{1}{k^+}\to\frac{\zeta}{k^+}$;
  - the lower limit becomes $\frac{k^+}{\zeta\vee}$;
  - the integration variable is renamed $\xi\to z$;
  - the phase becomes $e^{-ik\cdot(y'-y)/(\zeta z)}$.

  Then

$$B = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\frac{k^+}{\vee}}^1\frac{d\zeta}{\zeta^2}D_{h/g}(\zeta,\mu)\int_{\frac{k^+}{\zeta\vee}}^1\frac{dz}{z^2}P_{gg}(z)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\zeta z}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}.$$

In B put $\xi=\zeta z$ at fixed z:

- the phase $e^{-ik\cdot(y'-y)/(\zeta z)} = e^{-ik\cdot(y'-y)\frac{1}{\xi}}$;
- the measure $\frac{d\zeta}{\zeta^2}\frac{dz}{z^2} = \frac{d\xi}{\xi^2}\frac{dz}{z}$;
- $D_{h/g}(\zeta) = D_{h/g}(\frac{\xi}{z})$;
- the limits become $\frac{k^+}{\vee}\le\xi\le 1$, $\xi\le z\le 1$.

So $B=-A$. Numerically this was checked with the phase and the plus distribution, including the $\log(1-\frac{k^+}{\zeta\vee})$ and $\log(1-\xi)$ from the cut supports: rel. 1×10^{-12}. The pole is absorbed into $D_{h/g}$.

6 Step 5: Row IV after the subtraction, in your form

Eq. (3.2) holds for any weight that does not depend on ξ. So the subtracted Row IV can be written in your split; only $W\to W+\frac{2}{\epsilon}$ changes:

$$\begin{aligned}\frac{d^3N_{\rm IV}^{\overline{MS}}}{d^3k} = {}& \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-2\gamma_E-\log\frac{(y'-y)^2\mu_{\overline{MS}}^2}{4}\Big]\\ &\times\int_{\frac{k^+}{\vee}}^1 d\xi\Big[\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{\xi(1-\xi)}+\Big(\xi(1-\xi)+\frac{1-\xi}{\xi}\Big)\frac{e^{-ik\cdot(y'-y)\frac{1}{\xi}}}{\xi^2}\Big]\mathcal{O}\\ &+\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\\ &\times\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-2\gamma_E-\log\frac{(y'-y)^2\mu_{\overline{MS}}^2}{4}\Big]\mathcal{O}\end{aligned}$$

$$+\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+\!+\Lambda}} d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}$$
$$\times\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big]\mathcal{O}. \tag{6.1}$$

- For $d_\perp=2$, each $\big[-2\gamma_E-\log\frac{(y'-y)^2\mu^2}{4}\big]$ becomes $\big[-2\gamma_E-\log\frac{(y'-y)^2\mu^2}{4}-1\big]$, and the last line keeps $d_\perp=2$.
- The logarithm $-2\gamma_E-\log\frac{(y'-y)^2\mu^2}{4}=-\log\frac{(y'-y)^2\mu^2}{c_0^2}$, with $c_0=2e^{-\gamma_E}$, vanishes at $\mu=c_0/|y'-y|$, i.e. $\mu\sim|\mathbf{k}|$.

7 Step 6: DGLAP

From the μ independence of the cross section.

- In Eq. (6.1), $\frac{\partial}{\partial\log\mu^2}\big[-2\gamma_E-\log\frac{(y'-y)^2\mu^2}{4}\big]=-1$. So Row IV contributes -1 times Eq. (2.2) without $(-\frac{2}{\epsilon})$, i.e. by Eq. (3.2) -1 times the braces of Eq. (3.3).
- Group I Rows II and V 2b contribute their $\delta(1-\xi)$ coefficients with -1 (their poles also come as $-\frac{2}{\epsilon}-2\gamma_E-\log(\dots\mu^2)$).
- The running of $g^2(\mu)$ in LO gives $-\frac{\alpha_s}{2\pi}\frac{\beta_0}{2}$.
- By Sec. 4 the sum is $-P_{gg}$:

$$\frac{\partial}{\partial\log\mu^2}\frac{d^3N_g^{\overline{MS}}}{d^3k}=-\frac{\alpha_s}{2\pi}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}.$$

Fragment this with $\int\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta,\mu)$ at the gluon momentum $(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$, as for B. Add $\frac{\partial D}{\partial\log\mu^2}$ in Eq. (5.1). Then $\frac{\partial}{\partial\log\mu^2}\frac{d^3N_h}{d^3k}=0$ reads

$$\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\Big[\frac{\partial D_{h/g}(\xi,\mu)}{\partial\log\mu^2}-\frac{\alpha_s}{2\pi}\int_\xi^1\frac{dz}{z}P_{gg}(z)D_{h/g}\Big(\frac{\xi}{z},\mu\Big)\Big]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}=0. \tag{7.1}$$

This holds for every k^+, every $\mathbf{k}$, and every projectile and target. Different ξ correspond, through $e^{-ik\cdot(y'-y)\frac{1}{\xi}}$, to different parent momenta $\mathbf{k}/\xi$. So the bracket vanishes for each ξ:

$$\boxed{\mu^2\frac{\partial}{\partial\mu^2}D_{h/g}(\xi,\mu)=\frac{\alpha_s(\mu)}{2\pi}\int_\xi^1\frac{dz}{z}\,P_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu\Big),\qquad P_{gg}\text{ of Eq. (4.2).}} \tag{7.2}$$

The same from $D^{(0)}$. In $d_\perp=2-\epsilon$, $\frac{d\alpha_s}{d\log\mu^2}=-\frac{\epsilon}{2}\alpha_s+O(\alpha_s^2)$. So

$$\frac{\partial}{\partial\log\mu^2}\Big[\frac{\alpha_s}{2\pi}\frac{2}{\epsilon}\Big]=-\frac{\alpha_s}{2\pi}\qquad(\texttt{sympy}:$$

1), and $\frac{dD^{(0)}}{d\log\mu^2}=0$ in Eq. (5.2) gives Eq. (7.2).

The fragmentation function to first order.

$$D_{h/g}(\xi,\mu)=D_{h/g}(\xi,\mu_0)+\frac{\alpha_s}{2\pi}\log\frac{\mu^2}{\mu_0^2}\int_\xi^1\frac{dz}{z}P_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu_0\Big)+O(\alpha_s^2). \tag{7.3}$$

8 Checks

Script `user_ff_checks.py`, results in `user_ff_checks.json`. All numerical checks keep the complex phase.

check	result
your form, upper limit $\frac{k^+}{k^+\!+\Lambda}$, vs. the exact Row IV (9 cases)	rel. 2×10^{-14}
your form, upper limit 1: difference $\to -i\frac{\Lambda}{k^+}k\cdot(y'-y)e^{-ik\cdot(y'-y)}$	1.3×10^{-4} at $\Lambda/k^+=10^{-4}$
$\int\frac{d\xi}{\xi(1-\xi)}$ between the limits (`sympy`)	0
W at $d_\perp=2-\epsilon$, Eq. (2.1) (`sympy`)	0
$\frac{g^2N_c}{4\pi^2}-\frac{\alpha_s}{2\pi}2N_c$ (`sympy`)	0
your subtraction $\to$ standard plus, Eq. (3.2)	rel. 1×10^{-15}
$\delta(1-\xi)$ tally (`sympy`)	$\frac{11}{6}-\log(1-\xi_0)$
momentum sum rule (`sympy`)	$\frac{11N_c}{6}$
$B=-A$ with the phase, plus distribution and cut supports	rel. 1×10^{-12}
$\partial_{\log\mu^2}\big[\frac{\alpha_s}{2\pi}\frac{2}{\epsilon}\big]$ over $-\frac{\alpha_s}{2\pi}$ (`sympy`)	1